Question No 1 :-

```
Declare seats [50][50]
Declare rows, cols, i, j, k, mid, d
Declare s, r, rw, rowfree, total, found
Delare foundRow, start, ok

total = 0
found = 0
foundrow = -1
start = -1

//read rows
ok = 0
while ok == 0
    print "Enter number of rows (1-50): "
    read rows
    if rows >= 1 AND rows <= 50
        ok = 0

//read seats per row
ok = 0
while ok == 0
    print "Enter number of seats per row (1-50): "
    read cols
    if cols >= 1 AND cols <= 50
        ok = 1

//read seat status
print "Enter seat status (0 = available, 1 = booked): "
for i = 0 to rows - 1
    for j = 0 to cols - 1
```

Scanned with CamScanner

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ok=0
while ok==0
    read seats[i][j]
    if seats[i][j]==0 OR seats[i][j]==1
        ok=1
    else
        print "Invalid, enter 0 or 1: "

// display the seating plan
print a blank line
print 4 spaces
for j=0 to cols-1
    print j and a space
print a new line
for i=0 to rows-1
    print "R" and i, then a space
    for j=0 to cols-1
        if seats[i][j]==0
            print "_ "
        else
            print "X "
    print a new line
// free seats in each rows and in total
print "free seats per rows"
for i=0 to rows-1
    rowfree=0
    if seats[i][j]==0
        rowfree=rowfree+1
    print rowfree and a space
```

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    total = total * rowfree
print "Total free:" and total

//read group size
ok = 0
while ok == 0
    print "Enter number of customers
           (1 to cols): "
    read k
    if k >= 1 AND k <= cols
        ok = 1
    else
        print "Invalid group size."
//search for k consecutive seats,
  middle row first
mid = rows / 2
for (d = 0 to rows ; found == 0)
    for (s = 0 to 1 ; found == 0)
        if s == 0
            r = mid - d
        else
            if d == 0
                skip to next s
            r = mid + d
        if r < 0 OR r >= rows
            skip to next s
        run = 0
        for (j = 0 to cols-1 ; found == 0)
            if seats[r][j] == 0
```

print

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run = run + 1
if run == k
    found = 1
    foundrow = r
    start = j - k + 1
else
    run = 0

if found == 1
    for j = 0 start to start + k - 1
        seats [foundrow] [j] = 1
    total = total - k
    print "Group of k seated in Row foundRow
           seats start to start + k - 1"

// display the updated seating plan
print a blank line
print 4 spaces
for j = 0 to cols - 1)
    print j and a space
print a new line
for i = 0 rows + 1 to rows - 1
```

```
print ("R")
print (i)
print ("    ")
  for j=0 to cols-1
     if seats[i][j]==0
        print "-  "
     else
        print "X "
  print("\n")
print ("Remaing available seats:" total)
else
   print ("No k consecutive seats available")
End
```